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PuG Symposium Heidelberg 2026

Neuroscientific Perspectives on Personality

The problem of weak associations between
personality and neurocognitive measures

Weak associations between personality and neurocognitive measures

Outline

Introduction

- Problem outline

Technical issues

- Reliability
- Range restriction

Conceptual issues

- Brunswik asymmetry
- Strong vs. weak situations

Integration



For slides, literature and code of this presentation, see:

<https://github.com/alex-strobel/PuG-2026-Talk>

Weak associations between personality and neurocognitive measures

Introduction: Problem outline

Problem outline

— What is a weak association in the first place?

Cohens convention does not pass the empirical test for individual differences research (and many other psychology subdisciplines!), most effects seem to be around $r \sim .10$ -.30.

— Then why do others observe strong associations so often?

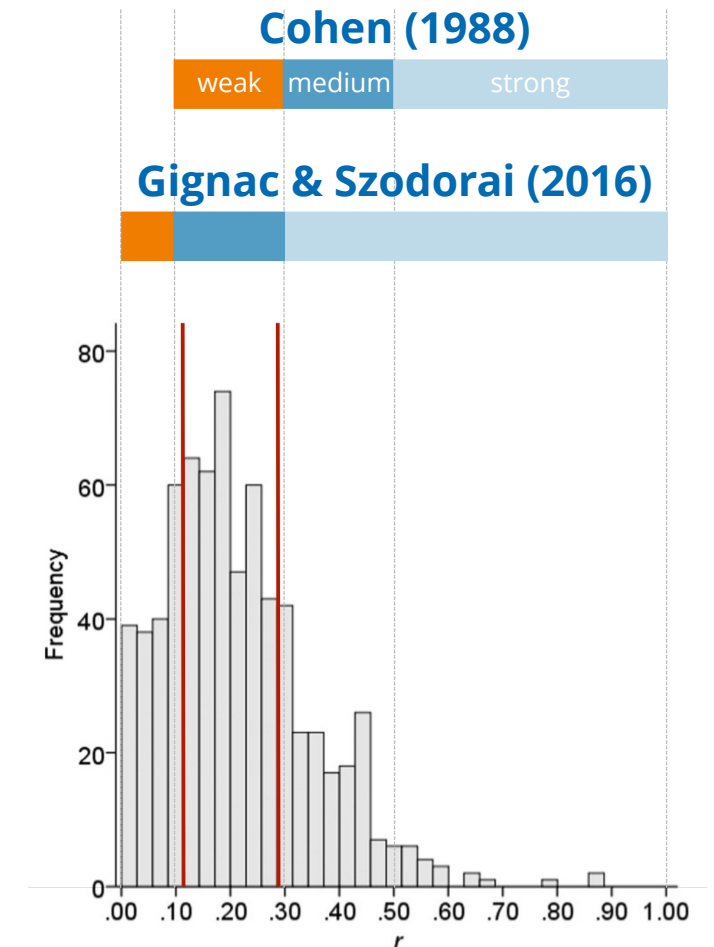
Biopsychological research is not likely to come up with stronger effects, but its effects *need to be strong* to show up in small samples (and to get published)

— But I don't find even any significant medium associations?

Power is maybe too low (required sample sizes to detect effects of $r = .10/.20/.30$ associations are $N = 780/193/85$ at a power of 80%)

— But even then, my effects are smaller as to be expected?

This may be a consequence of publication bias (see Open Science Collaboration, 2005), but also simply due to technical issues ...



Weak associations between personality and neurocognitive measures

Technical issues: Reliability

Correlation attenuation

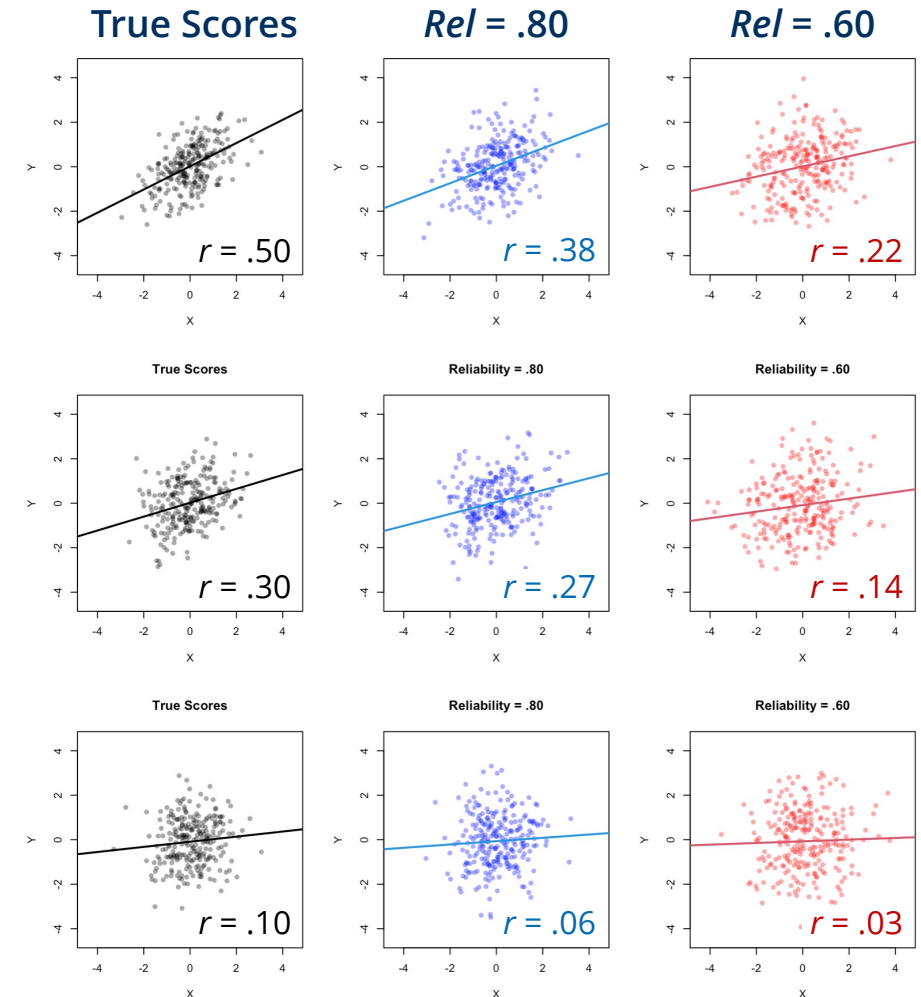
- the observed correlation between two variables is weaker than their "true" relationship due to unreliability in the data and/or measurement error

Suboptimal reliability is a pervasive issue

- Intelligence research: $\sim .90$ (own estimate)
- Personality research: $\leq .80$ (own estimate)
- Cognitive research: $\sim .60$ (Enkavi et al., 2019)
- Imaging research: $\sim .40$ (Elliott et al., 2020)

Consequences

- use high reliability measures / capture measurement error
- correct expected effect size for reliability of your measures for estimation of required sample size at a given α and $1-\beta$
- correct observed r for attenuation: $r_{\text{true}} = r_{\text{obs}} / \sqrt{\text{Rel}_X * \text{Rel}_Y}$



Weak associations between personality and neurocognitive measures

Technical issues: Range restriction

Correlation attenuation

- the observed correlation between two variables is weaker than their "true" relationship due to restriction of variance in the data

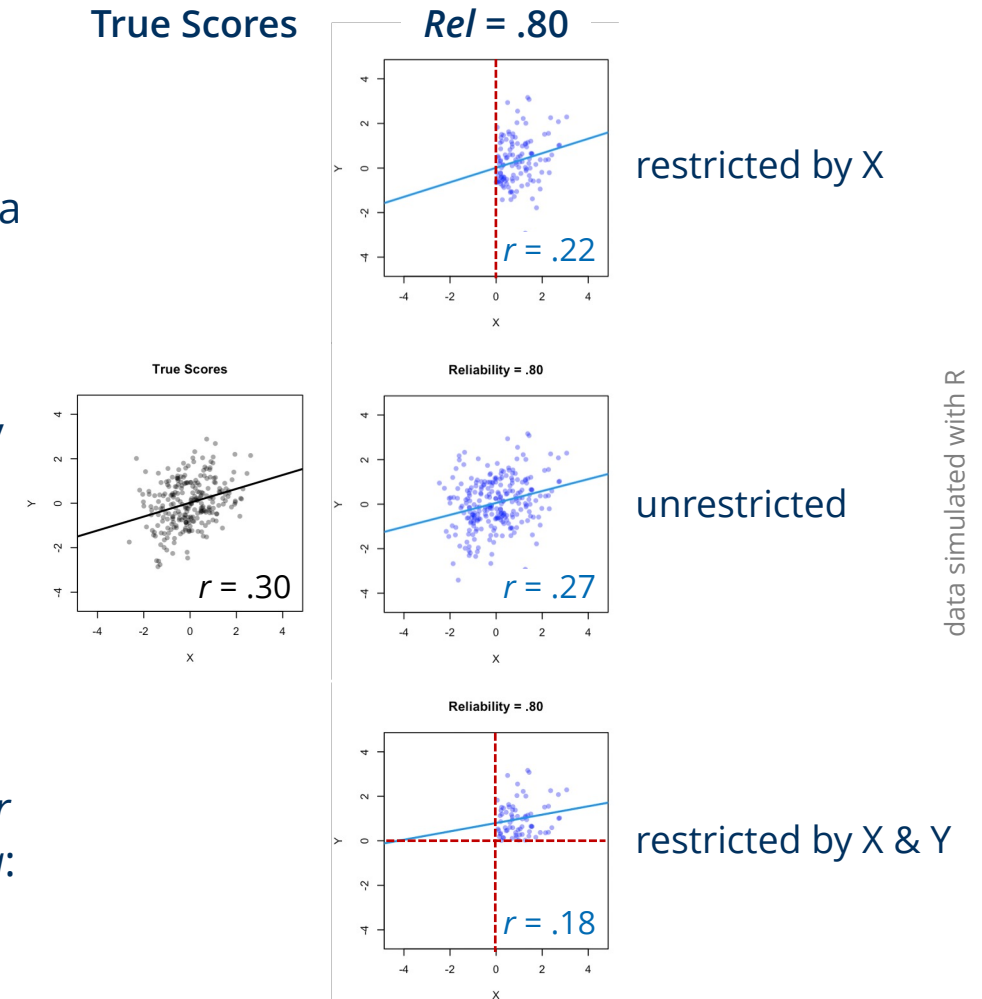
Range restriction is an issue in psychological research

- Participants often young and intellectually capable (and *WEIRD*)
- Fictitious example: $r_{true} = .30$ between age and working memory is $r_{obs} = .27$ due to reliability issues, and $r_{obs} = .22/.18$ due to range restriction (only young/young and capable individuals)

Consequences

- examine individuals from general population (**not** psychology plus architecture and engineering students ;-)
- if variances in general population are known, correct observed r for range restriction – easiest case, variance restriction only in a :

$$R_{ay} = r_{ay} \left(\frac{SD_a}{sd_a} \right) / \sqrt{(1 - r_{ay}^2) + r_{ay}^2 \left(\frac{SD_a^2}{sd_a^2} \right)}$$



Details and further correction methods: Carretta, T. R. & Ree, M. J. (2022). *Milit Psychol*, 34(5), 551–569.

<https://doi.org/10.1080/08995605.2021.2022067>

Weak associations between personality and neurocognitive measures

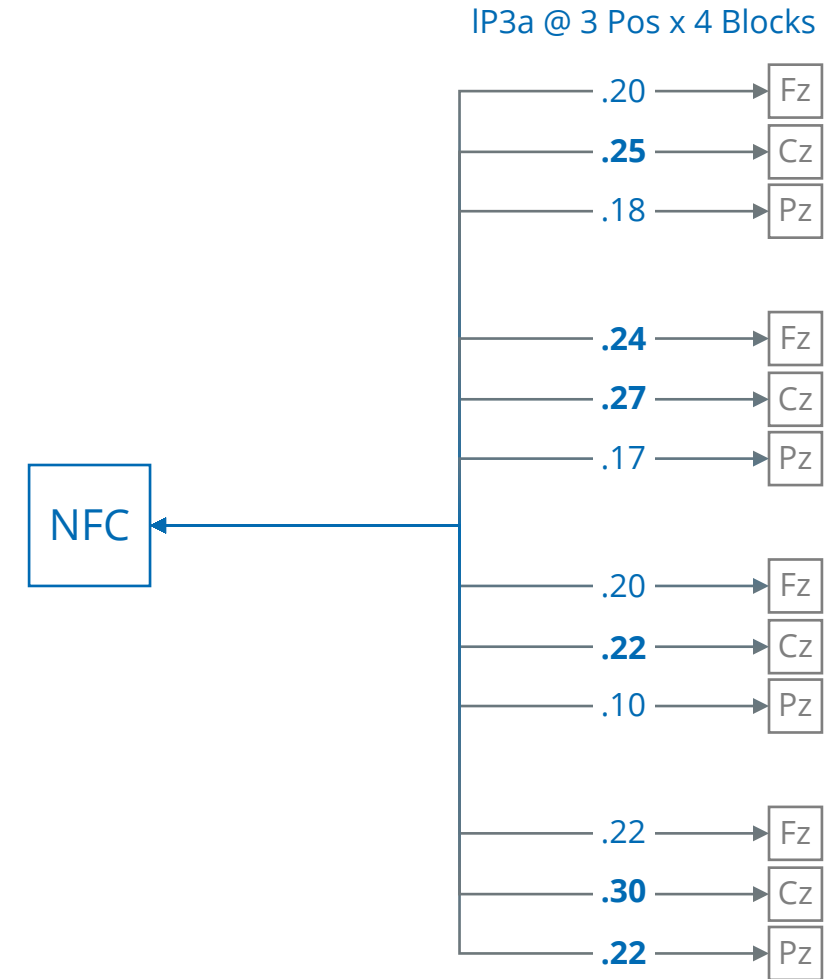
Conceptual issues: Brunswik asymmetry

Brunswik asymmetry (in terms of generality)

- correlations between a trait and a *specific* behavior are often smaller than those between the trait and more *general* behavior

Generality asymmetry is often a problem in psychology

- example here: we try to link a rather broad trait like Need for Cognition (NFC) to very specific EEG measures in a novelty oddball task (here: the late P3a in some block at some electrode)



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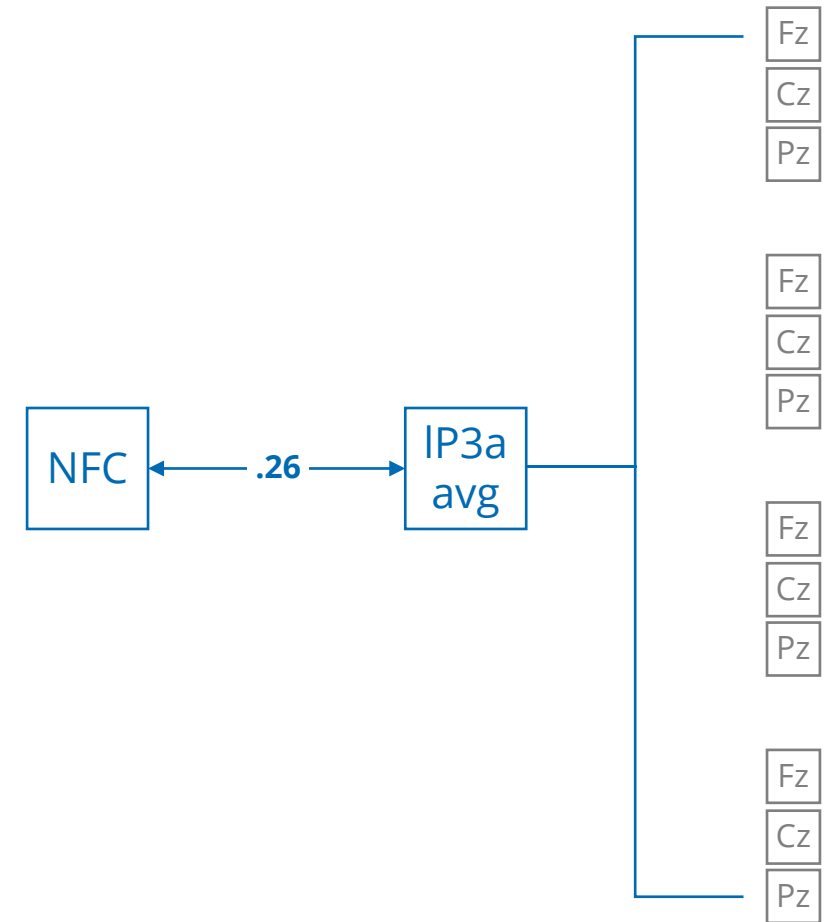
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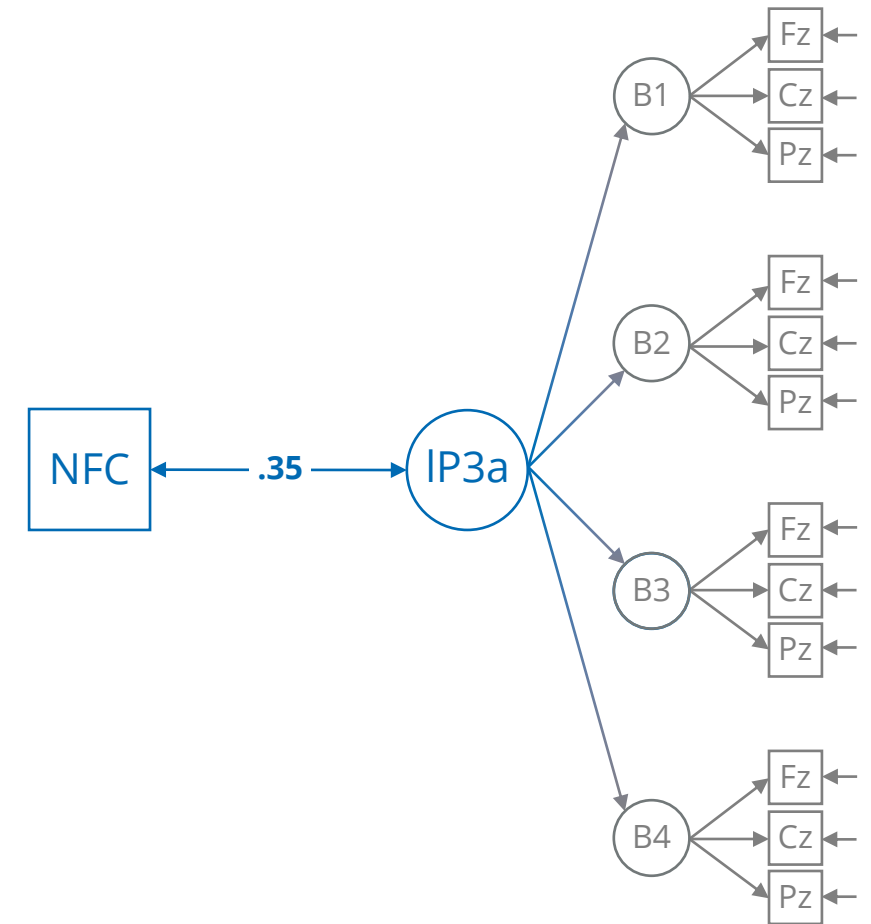
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Consequences

- try to associate constructs rather than specific measures via aggregation/structural equation modelling (and by doing so, also get rid of measurement error, see above)
- considerably more effort when using more than one measure of IV and DV; arriving at a proper SEM can be a p*** in the a**



Weak associations between personality and neurocognitive measures

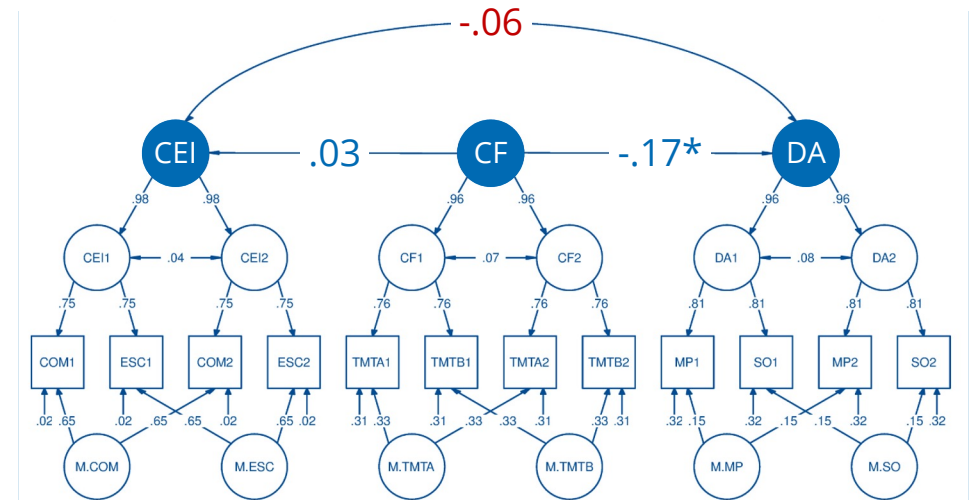
Conceptual issues: Strong vs. weak situations

Strong vs. weak situations

- cf. Mischel: strong situations minimize, weak situations maximize individual differences

Situational affordance is an issue in cognitive studies

- experimental paradigms often *designed* to minimize the influence of individual differences in order to maximize situational influence
- individuals with different trait levels tend to behave similar due to situation strength, resulting in weak associations



Strobel et al. (2020): in a sample of $N = 217$ university students, trait *Cognitive Effort Investment*, CEI, was not related to *Demand Avoidance*, DA (controlled for cognitive functioning, CF) likely because the task was optimized to demonstrate that DA is a general phenomenon

Weak associations between personality and neurocognitive measures

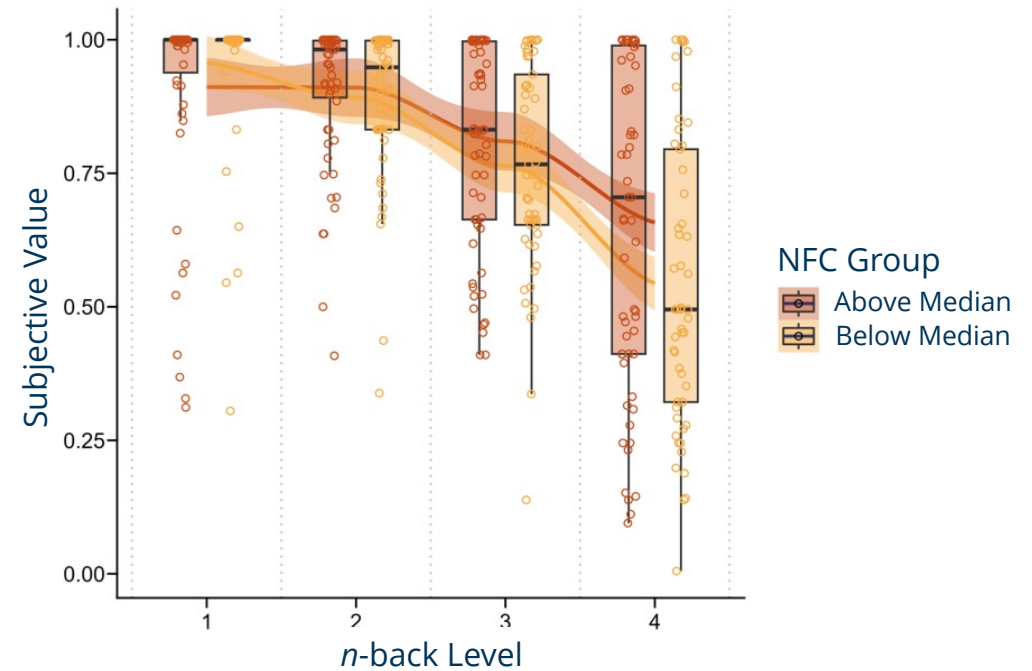
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Situational affordance is an issue in cognitive studies

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- individuals with different trait levels tend to behave similar due to situation strength, resulting in weak associations
- if situation strength is reduced like in the n -back task and individuals are asked for subjective values of different n -back levels, the expected nonlinear person $\times$ situation interaction associations shows up
- tasks like the Novelty Oddball or measurements under resting conditions also create rather weak situations



Zerna, Scheffel et al. (2023): in a sample of $N = 116$ university students, high vs. low NFC individuals showing expected patterns of subjective values over n -back levels

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Conceptual issues: Strong vs. weak situations

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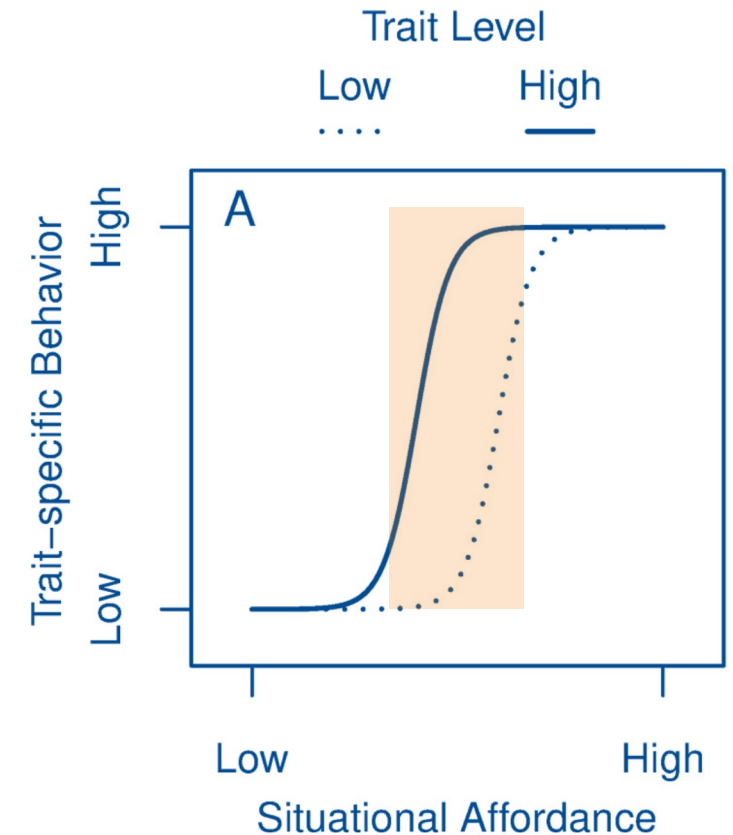
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Consequences

- establish experimental or ecological situations with medium situation strength so that individuals can behave in line with their dispositions



Weak associations between personality and neurocognitive measures

Integration: Summary and suggestions

Problem

in individual differences research, *medium* associations can be assumed at best, but they may appear *weak* because of ...

Unreliability of the measures used

— carefully select measures and/or measurement models to avoid and/or correct for attenuation

Variance restriction of the samples examined

— carefully select samples and/or know true variances of your measures to correct for restriction

Generality asymmetry of the measures employed

— make sure that X and Y are of the same generality level (aggregation/SEM, also captures error)

Strong vs. weak measurement situations

— establish experimental or ecological situations with medium situation strength so that individuals can behave in line with their dispositions

Suggested reading: DeYoung, C. G., Hilger, K., et al. (2025). *J Cogn Neurosci*, 37(6), 1023-1034.

https://doi.org/10.1162/jocn_a_02297

Thank you!

Details?

<https://github.com/alex-strobel/PuG-2026-Talk>

Questions?

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